

CROSSEDIT: CROSS-MODAL TRAINING ENABLES RICH AUDIO-VISUAL EDITING

William Chen^{1†,2*}, Prem Seetharaman^{1*}, Ke Chen^{1*}, Oriol Nieto^{1*},
 Kevin Duarte¹, Siddarth Srinivasan Iyer¹, Mamshad Nayeem Rizve¹, Zhiwen Cao¹
 Shinji Watanabe², Yuanjun Xiong¹, Jiaming Zhang¹, Zeyu Jin¹, Justin Salamon¹
 Adobe Research¹
 Carnegie Mellon University²
 williamchen@cmu.edu, seethara@adobe.com

ABSTRACT

Multimedia editing requires generative models to interpret complex, compositional instructions and modify only the desired elements across modalities, a capability largely beyond existing systems. Current approaches are typically limited to simple, single-attribute edits within one modality, since diverse, high-quality editing pairs are hard to obtain, especially for cross-modal tasks where aligned audiovisual (AV) data is scarce. We present **CrossEdit**, a unified omni-modal editing model for images, audio, and video that performs AV movie scene edits zero-shot through cross-modal transfer. Our key observation is that complex instruction-following, once learned in any modality, generalizes to others. We exploit the additive nature of acoustic signals to procedurally generate large-scale audio editing pairs with compositional instructions, and show that fine-tuning on this synthetic data and self-supervised AV masked reconstruction, alongside a targeted set of curated cross-modal tasks, induces instruction-following that transfers zero-shot to unseen modality and instruction combinations. To evaluate this capability, we release **CrossEditBench**, a human-annotated benchmark of AV edits on movie scenes, and propose **AV-FES**, a metric that jointly scores instruction following and consistency. We show that our proposed techniques improve performance on zero-shot AV movie scene editing, while maintaining or improving performance on lip-synced speech editing and standard image, video, and audio editing benchmarks. Demos are available at <https://wanchichen.github.io/crossedit/>.

1 INTRODUCTION

Recent advances in multi-modal generative models have enabled the synthesis of images, video, and audio from natural language descriptions (Wang et al., 2023a; Tian et al., 2026; Chen et al., 2026; Deng et al., 2025; Zhou et al., 2025; Shi et al., 2024; Evans et al., 2025; Hung et al., 2024), with rapidly improving fidelity and controllability. Yet generation is only half the story. In practice, it is unlikely that a generative model will produce exactly what a user desires in a single attempt — natural language is inherently ambiguous, and creative workflows are inherently iterative. Users will inevitably wish to modify some aspects of generated media while preserving others. This is most difficult for audio-visual (AV) content, where an edit such as adding a motorbike to a street scene must change both picture and sound, keep them synchronized, and leave everything else untouched.

Existing editing models, however, remain largely siloed by modality (Chen et al., 2026; Wang et al., 2023b; Ellis et al., 2025; Lan et al., 2025; Peng et al., 2024), and the few multi-modal systems support only simple, single-attribute edits (e.g., “make her hair blue”). The bottleneck is data: editing pairs are easy to construct for atomic, single-modality edits, but AV-to-AV (AV2AV) pairs with compositional instructions rarely occur naturally and are expensive to curate.

We present **CrossEdit**, an omni-modal editing model that can edit real movie scenes from free-form instructions (Figure 1) *zero-shot*, without ever being trained on such data. Our work rests on two

*Equal Contribution. † Work done while at Adobe.

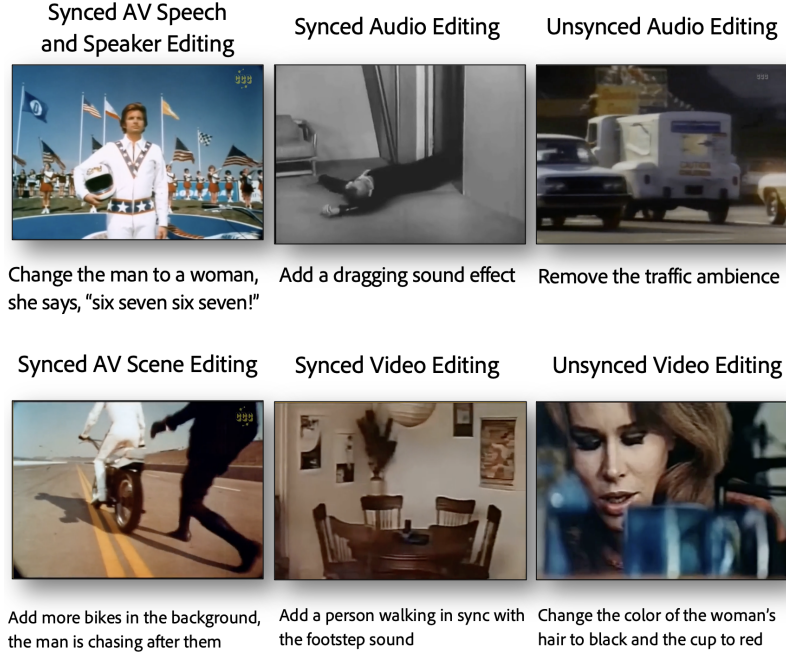


Figure 1: **Instruction types in CrossEditBench that CrossEdit must handle zero-shot.** Models must implicitly determine which modalities to edit and what to synchronize across audio and video.

hypotheses: (1) *complex instruction-following is modality-agnostic*, so once learned in one modality it transfers to others; and (2) *masked AV reconstruction can serve as a self-supervised AV2AV editing task*. We operationalize the first hypothesis by exploiting an asymmetry between auditory and visual signals. Acoustic signals are physically additive (mixing waveforms yields a valid audio scene), so large-scale pairs of complex instructions and simulated mixtures can be easily generated, whereas visual compositing lacks this physical fidelity. We hypothesize that these synthetic audio editing pairs, combined with a core set of curated cross-modal tasks (such as masked AV reconstruction), can better generalize complex instruction-following capabilities across modalities.

To evaluate zero-shot AV2AV movie scene editing, we introduce CrossEditBench, a human-annotated benchmark of edits on real movie scenes that require jointly reasoning over the instruction, video, and audio, and AV-FES, a novel metric that directly evaluates editing performance with a single interpretable score. Our results show that our proposed techniques allow the model to better perform AV2AV movie scene editing without any training data for the task, while maintaining, if not improving, performance on trained tasks such as video and image editing.

2 RELATED WORK

Multi-modal foundation models. Unifying images (Zhai et al., 2022; Ho et al., 2020; Song et al., 2021; Salimans & Ho, 2022), video (Chen et al., 2024a), audio (Radford et al., 2023; Pratap et al., 2023; Chen et al., 2024b; 2025b; Valle et al., 2025), and text (Brown et al., 2020; Jaech et al., 2024; Yang et al., 2025) within a single model has drawn significant attention (Team et al., 2025; Bai et al., 2025b; Chu et al., 2024; Tian et al., 2025b; Chen et al., 2026; Deng et al., 2025; Zhou et al., 2025). Most work pairs visual or audio generation with text (Tian et al., 2025b; Deng et al., 2025; Zhou et al., 2025; Tian et al., 2025a; 2026; Liao et al., 2025), and any-to-any models (Luo et al., 2026; Tang et al., 2024; Li et al., 2025) typically cannot output multiple modalities simultaneously. Seedance 2.0 (Seedance et al., 2026), LTX-2 (HaCohen et al., 2026), and UniAVGen (Zhang et al., 2025) achieve synchronized AV synthesis via dual-stream diffusion transformers, but can only *generate*, not *edit*.

Instruction-based media editing. Editing has been explored within each modality: for images, via synthetic editing triplets (Brooks et al., 2023; Zhang et al., 2023; Geng et al., 2024; Zhao et al., 2024; Wei et al., 2025b); for video, via temporal consistency mechanisms on image diffusion models (Wu

Table 1: **Training and Evaluation tasks for CrossEdit.** Unlike trained lip-synced speech editing, zero-shot speech + speaker edits also change who is speaking. [†] Voice cloning only.

Task	Example Instruction	Pre-train	SFT	Eval.
<i>Editing</i>				
General image editing (I2I)	“Make her hair blue”	✓	✓	\$5.3
General video editing (V2V)	“Turn the car red”	✓	✓	\$5.2
Speech editing [†] (A2A)	“Change the speech to ‘See you soon’”	✓	✓	–
Complex audio editing (A2A)	“Remove the music; make the rain louder”	✗	✓	\$D
Lip-synced speech editing (AV2AV)	“Change the speech to ‘We made it out’”	✓	✓	\$5.1
Masked inpainting (AV2AV)	“Inpaint this video”	✗	✓	–
<i>Generation (no AV input)</i>				
Text-to-X (T2I, T2V, T2A)	“A dog barking in the rain”	✓	✓	–
Animation ($\{I,A,IA\}2\{V,AV\}$)	“A person talking”	✓	✓	–
<i>Zero-shot tasks (all AV2AV)</i>				
Synced AV speech + speaker editing	“Change the man to a woman, she says...”	✗	✗	\$4
Synced audio editing	“Add a dragging sound effect”	✗	✗	\$4
Unsynced audio editing	“Remove the traffic ambience”	✗	✗	\$4
Synced AV scene editing	“Add more bikes in the background”	✗	✗	\$4
Synced video editing	“Add a person walking to the footsteps”	✗	✗	\$4
Unsynced video editing	“Change the woman’s hair to black”	✗	✗	\$4

et al., 2023a; Geyer et al., 2023; Qi et al., 2023) and, more recently, large editing datasets (Wu et al., 2025c; Bai et al., 2025a; He et al., 2025; Wei et al., 2025a; Ju et al., 2026); and for audio, via mixtures of real-world signals for a preset list of supported operations (Wang et al., 2023b; Ellis et al., 2025; Huang et al., 2023), later extended with LLM-generated instructions (Lan et al., 2025; Chen et al., 2026). Early AV2AV works relied on cascaded pipelines (Ishii et al., 2025; Polyak et al., 2024; Zheng et al., 2025) or sampling tricks applied to uni-modal models (Lin et al., 2026). End-to-end AV editing has only emerged recently, but concurrent works target narrower settings. For example, SpongeBob (Liang et al., 2026) and InstructAV2AV (Zheng et al., 2026) only perform simpler instructions that are doable via inpainting, such as addition and removal. Spongebob requires per-modality captions, an object mask, and a reference image, so the user rather than the model decides what changes in each modality; its code, weights, and benchmark are also unreleased. Both train on paired AV editing data for their target tasks and are AV-only, whereas CrossEdit is omni-modal and edits real movie scenes from free-form instructions zero-shot. Most recently, MiniMax H3 (MiniMax, 2026) unifies generation and editing of images, audio, and video within a single model. To our knowledge, CrossEdit is the first work to show that instruction-based AV2AV scene editing can be acquired zero-shot through cross-modal transfer of instruction following.

3 TRAINING FOR CROSS-MODAL TRANSFER

Built on a shared backbone (Section 3.1), CrossEdit acquires zero-shot AV scene editing from three data sources: mined lip-synced speech editing pairs (Section 3.2), synthetic audio editing pairs with complex instructions (Section 3.3), and AV masked reconstruction (Section 3.4).

3.1 ARCHITECTURE

Let $\mathcal{M} = \{\mathcal{I}, \mathcal{A}, \mathcal{V}\}$ denote the image ($\mathcal{I} = \mathbb{R}^{H \times W \times 3}$), audio ($\mathcal{A} = \mathbb{R}^{T_a \times 2}$), and video ($\mathcal{V} = \mathbb{R}^{T_v \times H \times W \times 3}$) spaces. We model omni-modal editing as $p_\theta(\hat{Y} \mid Y, X)$, where X is a natural language instruction, and $Y = (y_1, \dots, y_n)$ and $\hat{Y} = (\hat{y}_1, \dots, \hat{y}_k)$ are input and output sequences whose elements each belong to some modality $m \in \mathcal{M}$, with their number and modalities varying across examples. We parameterize p_θ with flow matching (Lipman et al., 2023) on latent representations of $\hat{Y}$. Using the linear interpolant $\hat{Y}_t = (1 - t)\hat{Y}_0 + t\epsilon$, where $\hat{Y}_0$ is the clean target, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, and $t \in [0, 1]$, we train a velocity field v_θ to regress $\epsilon - \hat{Y}_0$:

$$\mathcal{L} = \mathbb{E}_{(Y, X, \hat{Y}_0), \epsilon, t} \left[\left\| v_\theta(\hat{Y}_t, t, Y, X) - (\epsilon - \hat{Y}_0) \right\|^2 \right]. \quad (1)$$

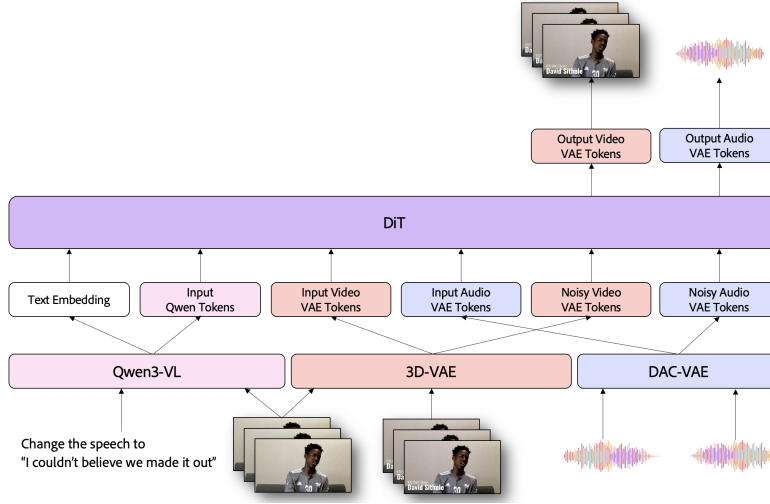


Figure 2: **Architecture of CrossEdit when encoding AV data.** For simplicity, we omit images, which are treated as a special case of single-frame video.

We independently replace Y and X with a null token $\emptyset$ at random during training, enabling classifier-free guidance (CFG) (Ho & Salimans, 2021) (Appendix B.3). At inference, we initialize $\hat{Y}_1 \sim \mathcal{N}(0, \mathbf{I})$ and integrate from $t = 1$ to $t = 0$ with an ODE solver.

In practice, p_θ is a DiT (Peebles & Xie, 2023) MoE (Shazeer et al., 2017; Peng et al., 2026) (3B active, 12B total parameters). Pre-trained VAEs (Kingma & Welling, 2014; Polyak et al., 2024; Kumar et al., 2023) encode audio (48kHz stereo $\rightarrow$ 256-dim. at 40Hz) and visuals (images and video). The visual VAE supports variable resolutions, mapping e.g. 512px to 48-dim. 32×32 latents at a fixed framerate (24 FPS $\rightarrow$ 6 FPS). Following Wu et al. (2025a), text, images, and video also receive clean semantic embeddings from a pre-trained Qwen3-VL (Bai et al., 2025b).

3.2 MINING LIP-SYNCD SPEECH EDITING PAIRS

To teach lip-synced speech editing, we mine pairs in which the same on-screen speaker says different sentences from in-the-wild, commercially usable or licensed videos. These cover only speech content edits, not scene edits (Section 3.5). For each video, we (1) remove non-speech audio with a language-queried source separation model (Liu et al., 2022; Wen et al., 2026); (2) segment and transcribe it with a speaker diarization model similar to Bredin et al. (2020), yielding transcripts, speaker labels, and time segments; (3) run face detection and tracking (Deng et al., 2020), discarding segments where a face is missing or goes off-screen; and (4) group retained segments by speaker to form before/after pairs. Filtering discards $\sim 96\%$ of the source data, but since each speaker yields many pairs, the overall yield is $\sim 20\%$ (e.g., 100K clips yield 20K pairs).

3.3 AUDIO EDITING DATA SYNTHESIS

Mined data alone is small, biased toward clean speech, and often inconsistent between input and output (e.g., differing background noise). We therefore complement it with synthetic audio editing data from two sources: (1) random mixtures of 2–4 audio files, each from one of four classes (speech, music, background, sound effect), following Kumar et al. (2026), which give acoustically plausible but semantically incoherent scenes of real audio; and (2) scenes synthesized by a tool-calling LLM equipped with generative audio models, following Chen et al. (2026), which yield more natural audio scenes that are more similar to that found in the real world, but with synthetically generated audio. Combining both sources yields a large-scale dataset of compositional instructions that overcomes the weaknesses of the mined data and helps the model generalize to unseen acoustic scenes; as we show in Section 4, the instruction following it teaches also transfers to AV movie scene editing.

3.4 AUDIO-VISUAL MASKED RECONSTRUCTION

While mined and synthetic pairs teach instruction following, neither requires the model to generate new visual content within an existing AV scene, and we find them insufficient for complex scene edits (Figure 1). Inspired by video generation (Voleti et al., 2022) and cross-modal representation learning

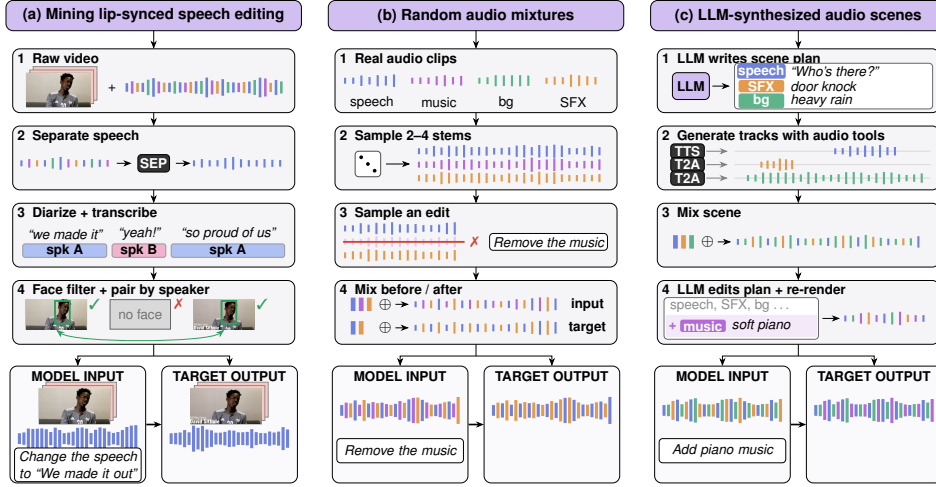


Figure 3: **Editing data sources.** (a) Our pipeline for mining lip-synched speech editing pairs. (b) Random audio mixtures and (c) Synthetic audio scenes provide audio editing pairs.

(Shi et al., 2022; Hsu & Shi, 2022; Gong et al., 2022), we frame AV masking as a self-supervised AV2AV editing task (Figure 4). Each AV sample receives one of three maskings: *temporal*, masking a random span of frames (and, 50% of the time, its audio); *spatial*, applying $s \sim \mathcal{U}\{0, S\}$ masks of 200×200 px to all frames, with $S=12$ for our 540p videos; or *spatio-temporal*, applying both. The model must reconstruct the unmasked clip given the instruction “inpaint this video”.

3.5 TRAINING

Given the scarcity of editing data, we start from an internal DiT checkpoint pre-trained on the tasks in Table 1, which we call **CrossEdit-Base**¹. CrossEdit-Base is a performant omni-modal editing model on its own, but it cannot consistently perform our target task of AV2AV movie scene editing. We add the synthetic audio editing and masked reconstruction data and fine-tune for 10K steps to obtain **CrossEdit** (data details, training settings, and inference hyperparameters in Appendix B.2).

Our central goal is to perform AV2AV editing of complex movie scenes without any AV2AV training data for those types of instructions. Table 1 lists every task CrossEdit sees during pre-training (aka CrossEdit-Base) and supervised fine-tuning (SFT). In all of its training, CrossEdit sees only two AV2AV instruction types: lip-synched speech editing (Section 3.2) and masked inpainting (Section 3.4). The instructions in Figure 1 (adding objects, removing ambience, changing speaker attributes) appear only in single-modality editing data (I2I, V2V, A2A), so AV2AV editing with them is *zero-shot*: it requires transferring instruction following learned in other modalities. Section 4 evaluates these zero-shot AV edits, and Section 5 checks that our techniques maintain or improve trained-task performance.

4 ZERO-SHOT AUDIO-VISUAL SCENE EDITING

We now evaluate our central claim that cross-modal training enables AV movie scene editing without AV scene editing data, using our proposed benchmark (Section 4.1) and metric (Section 4.2).

4.1 CROSSEDITBENCH

As no public benchmark exists for instruction-based AV movie scene editing, we release **CrossEdit-Bench**: 100 clips from the evaluation split of OSSL (Kim et al., 2025), which is built from public domain movies, each annotated by humans with an edit instruction. Annotators also specify what should change in each modality (e.g., “add a cow and remove the music” → audio: [add cow mooing, remove music]; video: [add cow]), which we release for future fine-grained evaluation.

¹Pre-training includes the mined lip-synched speech editing pairs (Section 3.2) but not the synthetic audio data (Section 3.3) or masked reconstruction (Section 3.4).

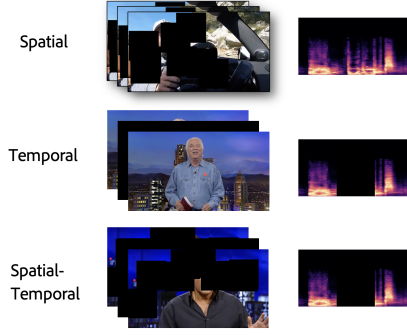


Figure 4: **AV masked reconstruction schemes.** All use the instruction “Inpaint this video”.

Instructions vary along two axes (Figure 1): *which modality the edit targets* and *whether it must be synchronized across modalities*. An *unsynced* edit must leave the other modality untouched (removing traffic ambience must not alter the video), while a *synced* edit must align with it (an added dragging sound must start and stop with the on-screen motion). This yields six categories (Table 1): unsynced audio, synced audio, unsynced video, synced video, synced AV scene, and synced AV speech and speaker editing. As none of them appear in AV2AV training data (Section 3.5), CrossEditBench directly measures zero-shot transfer. More examples are in Appendix C.1.

4.2 AV-FES: AUDIO-VISUAL FAITHFUL EDIT SCORE

Existing automatic metrics for AV editing score the edited output in isolation from the input or instruction and often remain disjoint across modalities (Girdhar et al., 2023; Lin et al., 2026; Zheng et al., 2025; Liang et al., 2026). Editing, as a highly conditional task, requires comparing the output against both the instruction and the input. AV-FES is grounded in two fundamental sub-scores often used in uni-modal editing evaluations, Instruction Following and Consistency (Chen et al., 2026; Guo et al., 2025; Ma et al., 2026), with respect to the text instruction and input media. While both are often evaluated on a 5-point Likert scale by a judge, such as a human annotator or LLM, the two axes pull in opposite directions. Leaving the input untouched maximizes consistency, while re-synthesizing the whole clip is the easiest way to satisfy the instruction. This makes it difficult to evaluate editing systems: how does one weigh gains in one sub-score against losses in the other?

We observed that this relationship is similar to the sub-scores of the commonly used F_1 metric: Precision (akin to Consistency) and Recall (parallel to Instruction Following). Motivated by this analogy, we therefore combine the two judge axes into a single score, the Audio-Visual Faithful Edit Score (AV-FES), computed separately for the video (V-FES) and audio (A-FES) modalities and averaged. For each input clip Y with instruction X , an LLM judge rates a system output $\hat{Y}$ on instruction following with respect to X , $\text{Ins}_m(\hat{Y}) \in \{1, \dots, 5\}$, and consistency with respect to Y , $\text{Cons}_m(\hat{Y}) \in \{1, \dots, 5\}$, for each modality $m \in \{\mathcal{V}, \mathcal{A}\}$. We also rate the unedited input Y itself under the same instruction, yielding $\text{Ins}_m(Y)$: the score a system obtains by doing nothing.

Instruction Gain. We first calculate the gain $g_m(\hat{Y})$, which quantifies the amount of achievable improvement the system obtained over the unedited input in Instruction Following:

$$g_m(\hat{Y}) = \text{clip}_{[0,1]} \left(\frac{\text{Ins}_m(\hat{Y}) - \text{Ins}_m(Y)}{5 - \text{Ins}_m(Y)} \right), \quad (2)$$

so $g_m = 0$ for any output the judge rates no better than the input (copying), and $g_m = 1$ when the output reaches the maximum score. Normalizing by the per-clip headroom $5 - \text{Ins}_m(Y)$ makes clips comparable regardless of how much the judge credits the untouched input.

Consistency. To make the sub-scores comparable, we then map the Consistency rating to the unit interval, $c_m(\hat{Y}) = \frac{\text{Cons}_m(\hat{Y}) - 1}{4}$, so that ratings of 5 and 1 map to 1 and 0.

Table 2: **CrossEditBench AV zero-shot scene editing results.** Gray columns are the modality-specific FES subscores. Improvements due to our proposed techniques are underlined.

		Overall	Visual subscores				Audio subscores				Alignment
Type		AV-FES $\uparrow$	V-FES $\uparrow$	Gain	Ins.	Cons.	A-FES $\uparrow$	Gain	Ins.	Cons.	ImageBind $\uparrow$
Input Video		—	—	—	2.81	3.97	—	—	2.19	3.95	19.79
LucyEdit + HY-XL	cascade	0.16	0.12	0.12	2.88	3.76	0.19	0.25	2.79	2.27	9.81
LucyEdit + HY-XXL	cascade	0.16	0.12	0.12	2.88	3.76	0.20	0.29	2.89	2.15	13.20
VACE + Coherent	cascade	0.19	0.12	0.14	3.16	3.44	0.26	0.34	3.12	2.49	13.22
AvED	sampling	0.12	0.09	0.10	2.81	3.20	0.14	0.13	2.49	3.93	19.38
AVI-EDIT	agentic	0.17	0.08	0.10	2.42	2.35	0.25	0.27	2.93	3.65	15.45
InstructAV2AV	e2e	0.20	0.15	0.15	2.85	3.03	0.24	0.26	2.85	3.20	16.08
MiniMax H3	e2e	0.23	0.24	0.23	3.14	3.38	0.22	0.22	2.77	3.70	18.90
CrossEdit-Base	e2e	0.26	0.20	0.24	2.88	2.00	0.31	0.39	3.15	3.52	14.30
CrossEdit	e2e	0.30	0.26	0.28	3.59	2.81	0.33	0.35	3.70	3.84	16.29

Faithful Edit Score. The per-modality FES is the harmonic mean of gain and consistency:

$$\text{FES}_m(\hat{Y}) = \frac{2 g_m(\hat{Y}) c_m(\hat{Y})}{g_m(\hat{Y}) + c_m(\hat{Y})}, \quad (3)$$

with $\text{FES}_m(\hat{Y}) = 0$ when $g_m(\hat{Y}) = c_m(\hat{Y}) = 0$. A straight-through copy of the input would score near 0 since $g_m \approx 0$, whereas a full re-synthesis that happens to satisfy the instruction is dragged down by a low c_m . With $\text{V-FES} = \text{FES}_V$ and $\text{A-FES} = \text{FES}_A$, the overall score for AV editing is their average:

$$\text{AV-FES}(\hat{Y}) = \frac{1}{2} \left(\text{FES}_V(\hat{Y}) + \text{FES}_A(\hat{Y}) \right). \quad (4)$$

All scores are computed per clip. AV-FES therefore rewards edits for each modality individually while penalizing unintended changes within a single metric. Additional details are in Appendix C.3.

4.3 RESULTS

Setup: We evaluate CrossEdit’s ability to edit entire AV scenes in a zero-shot manner (Section 3.5) using CrossEditBench (Section 4.1). We report AV-FES using Gemini 3.1 Pro (Section 4.2) and AV alignment via ImageBind score (Girdhar et al., 2023). We compare to two concurrent end-to-end models, (InstructAV2AV (Zheng et al., 2026) and the production-grade, 33B-parameter MiniMax H3 (MiniMax, 2026)), modular cascades (Lucy Edit (Decart AI, 2025) (V2V) + HY-Video-Foley XL/XXL (Shan et al., 2025) (V2AV) and VACE (V2V) + Coherent (VA2A) (Jiang et al., 2025; Ishii et al., 2025)), an agentic system (AVI-EDIT (Zheng et al., 2025)), and zero-shot delta denoising (AvED (Lin et al., 2026)). Subjective test details are in Appendix C.2.

Results: CrossEdit achieves the highest AV-FES, V-FES, and A-FES, driven by the strongest instruction following in both modalities (Table 2). It outperforms both concurrent end-to-end models, including MiniMax H3 (AV-FES 0.30 vs. 0.23) despite being under half its size (12B vs. 33B parameters), which is competitive on video (V-FES 0.24 vs. 0.26) but under-edits the audio (audio instruction following 2.77 vs. 3.70); InstructAV2AV trails on both modalities (AV-FES 0.20). Cascades lack cross-modal context, so the audio stage re-synthesizes the soundtrack and audio consistency collapses. AvED stays close to the input, attaining the highest ImageBind score but among the lowest gains, while AVI-EDIT degrades the video. CrossEdit also improves over CrossEdit-Base on every FES score (Section 4.4). Finally, our conclusions are robust to the choice of judge: Qwen3-Omni (Xu et al., 2025) correlates with Gemini at 0.90 (Pearson and Spearman) on system-level AV-FES, and both judges’ rankings closely match human ratings (AV-FES Pearson 0.92 and 0.96), albeit over only 4–5 shared systems (Appendix C.4). Example outputs are in Appendix C.1.

4.4 ABLATION OF TRAINING METHODS

Table 3 reports human ratings, and the FES computed from them, as each training component is added. CrossEdit-Base attempts edits but often alters unrelated content, giving the lowest visual consistency of all systems. Adding synthetic audio editing data, which contains no visual content, improves instruction following in both modalities as well as visual consistency, raising AV-FES from 0.26 to 0.35, consistent with cross-modal transfer. AV masked reconstruction further improves all

Table 3: **Ablation on zero-shot AV scene editing with human ratings** (344 per system; Ins. and Cons. with 95% CIs). Improvements over CrossEdit-Base are underlined.

	Overall	Visual subscores				Audio subscores			
	AV-FES $\uparrow$	V-FES $\uparrow$	Gain	Ins.	Cons.	A-FES $\uparrow$	Gain	Ins.	Cons.
Input	—	—	—	2.75 $\pm$ 0.24	4.71 $\pm$ 0.11	—	—	2.29 $\pm$ 0.25	4.68 $\pm$ 0.11
LucyEdit + HY-XL	0.11	0.14	0.12	2.61 $\pm$ 0.25	3.93 $\pm$ 0.20	<u>0.08</u>	0.16	2.17 $\pm$ 0.20	1.74 $\pm$ 0.20
LucyEdit + HY-XXL	0.14	0.17	0.15	2.74 $\pm$ 0.21	3.58 $\pm$ 0.20	<u>0.12</u>	0.17	2.22 $\pm$ 0.20	1.95 $\pm$ 0.21
CrossEdit-Base	0.26	0.25	0.34	2.85 $\pm$ 0.20	2.37 $\pm$ 0.19	<u>0.27</u>	0.32	2.89 $\pm$ 0.21	2.80 $\pm$ 0.21
+ Audio Editing	<u>0.35</u>	<u>0.35</u>	0.43	3.31 $\pm$ 0.21	2.94 $\pm$ 0.18	<u>0.35</u>	0.46	3.25 $\pm$ 0.21	2.81 $\pm$ 0.19
+ AV Masked Reconstruction (CrossEdit)	0.43	0.42	0.47	3.48 $\pm$ 0.18	3.21 $\pm$ 0.18	0.44	0.47	3.44 $\pm$ 0.18	3.44 $\pm$ 0.20

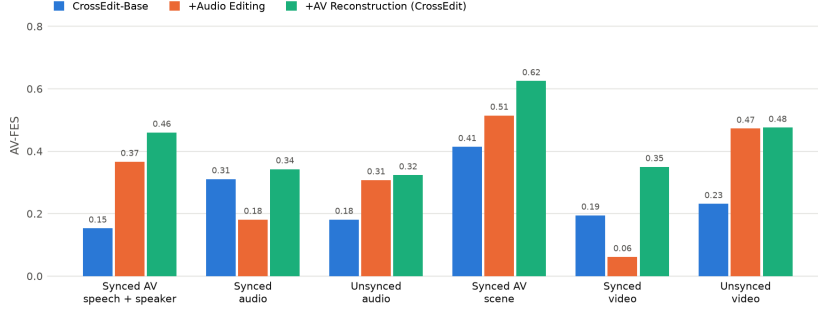


Figure 5: **Human-rated AV-FES on CrossEditBench, by instruction category.**

four axes, reaching 0.43. The cascades best preserve the video but barely follow visual instructions and overwrite the original soundtrack (AV-FES 0.11–0.14). On lip-synced speech editing, audio editing data likewise improves visual consistency (Appendix C.2).

Figure 5 breaks down human-rated AV-FES by type. CrossEdit improves over CrossEdit-Base in every category, with the largest gains on synced AV scene edits (0.41 to 0.62), speech+speaker edits (0.15 to 0.46), and unsynced video edits (0.23 to 0.48). Audio editing data alone helps most on edits that mainly require following the instruction, but hurts synced audio and synced video edits (0.31 to 0.18 and 0.19 to 0.06), where the new content must be generated in time with the other modality. Adding AV masked reconstruction recovers and surpasses CrossEdit-Base on both (0.34 and 0.35), consistent with its role of teaching the model to generate content that fits the existing scene.

5 RETAINING OMNI-MODAL PERFORMANCE

Having shown that cross-modal training enables zero-shot AV scene editing, we verify that it does not come at the expense of the lip-synced speech editing, image editing, and video editing tasks CrossEdit was trained on. We note that we primarily focus on the comparison between CrossEdit and CrossEdit-Base, with other models shown for reference. Results on audio editing are in Appendix D.

5.1 LIP-SYNCHED SPEECH EDITING

Setup: We use an internal set of 600 IMDB celebrity interviews. Given a new spoken sentence, the model must regenerate the audio and video with correspondingly synchronized lip movements and facial expressions (Figure 2). We measure WER and WavLM (Chen et al., 2022) speaker similarity (SPK) to the input; DINO (Oquab et al., 2024) similarity to the input and across frames; AV sync via LSE-D and LSE-C (Chung & Zisserman, 2017); and human ratings (Appendix C.2). As no public end-to-end system exists, we compare to cascades of F5-TTS (Chen et al., 2025c) with HuMo (Chen et al., 2025a) or StableAvatar (Tu et al., 2025), which animate the first input frame.

Results: Our SFT improves speaker similarity, video input consistency, and LSE-D, with minor gains in temporal consistency (Table 4). F5-TTS remains stronger on WER and SPK as a dedicated TTS system, but the cascades cannot preserve the input video, and human raters strongly prefer both CrossEdit models, especially for video consistency. Qualitatively, SFT mainly improves preservation of the input’s visual background (Figure 9 in Appendix C.6), reflected in higher video consistency.

Table 4: **Lip-synced speech editing results.** Improvements due to our techniques are underlined.

	Speech		Video		Sync		Human Ratings			
	WER ↓	SPK ↑	Time ↑	Input ↑	LSE-D ↓	LSE-C ↑	Audio		Video	
	Ins. ↑	Cons. ↑	Ins. ↑	Cons. ↑						
Input	-	-	-	-	-	-	3.29	4.39	3.29	4.50
F5-TTS	7.5	89.3	-	-	-	-	-	-	-	-
+ HuMo	-	-	99.1	67.4	9.12	5.88	3.30	2.97	2.60	2.10
+ StableAvatar	-	-	97.5	42.0	12.45	2.13	3.21	2.88	2.17	1.78
CrossEdit-Base	10.3	86.9	99.1	90.0	9.00	5.87	3.98	3.81	4.05	4.22
CrossEdit	10.9	<u>88.5</u>	<u>99.2</u>	<u>92.1</u>	<u>8.92</u>	5.64	3.96	<u>3.82</u>	4.02	<u>4.42</u>

Our ablation (Table 8) attributes most of this gain to the audio editing data: we hypothesize that simulated audio editing pairs teach the model to edit only what is necessary.

5.2 VIDEO EDITING

Setup: We evaluate on EditVerseBench (Ju et al., 2026) with GPT-4o-judged editing quality (Hurst et al., 2024), PickScore, text alignment, and CLIP/DINO temporal consistency. Our goal is to retain CrossEdit-Base’s performance: dedicated video editing models (InsV2V (Cheng et al., 2024), Lucy Edit (Decart AI, 2025), and EditVerse (Ju et al., 2026)) are included as reference points.

Results: Rather than merely retaining video editing performance, our training improves 5 of 6 metrics over CrossEdit-Base (Table 5), and CrossEdit remains in the range of dedicated models such as Lucy Edit. We hypothesize that the synthetic audio data exposes the model to more diverse instructions.

Table 5: **Video editing on EditVerseBench.** Improvements from our techniques are underlined.

	VLM Judge	Quality	Text Align.		Temporal Cons.	
	Editing Quality ↑	Pick Score ↑	Frame ↑	Video ↑	CLIP ↑	DINO ↑
InsV2V	5.21	19.39	24.99	22.54	97.15	96.57
Lucy Edit	5.89	19.67	26.00	23.11	98.49	98.38
EditVerse	7.65	20.07	26.73	23.93	98.56	98.40
CrossEdit-Base	6.86	19.42	23.10	20.10	98.00	96.90
CrossEdit	6.85	<u>19.82</u>	<u>25.21</u>	<u>22.26</u>	<u>98.43</u>	<u>97.52</u>

5.3 IMAGE EDITING

Setup: We use ImgEdit (Ye et al., 2026), which scores 9 editing categories with a GPT-4o judge. As with video, our goal is to retain CrossEdit-Base’s performance; open SOTA image editing models (Wu et al., 2025b;a; Lin et al., 2025; Deng et al., 2025) are included only as reference points.

Results: Adding audio data and AV masked reconstruction slightly improves the overall score over CrossEdit-Base (3.69 to 3.72), with large gains in the Action, Adjust, Hybrid, and Remove categories (Table 6). Both remain competitive with dedicated image models of similar or larger size.

Table 6: **Image editing results on ImgEdit (1–5, higher is better).** Dedicated image editing models are shown for reference, with scores from Wu et al. (2025a)

	Param.	Add	Adjust	Extract	Replace	Remove	Background	Style	Hybrid	Action	Overall
BAGEL	14B	3.56	3.31	1.70	3.30	2.62	3.24	4.49	2.38	4.17	3.20
UniWorld-V1	12B	3.82	3.64	2.27	3.47	3.24	2.99	4.21	2.96	2.74	3.26
OmniGen2	7B	3.57	3.06	1.77	3.74	3.20	3.57	4.81	2.52	4.68	3.44
Qwen-Image	20B	4.38	4.16	3.43	4.66	4.14	4.38	4.81	3.82	4.69	4.27
CrossEdit-Base	12B	4.37	4.06	1.48	3.61	3.43	3.88	4.54	3.22	4.48	3.69
CrossEdit	12B	4.20	<u>4.29</u>	<u>1.74</u>	3.61	<u>4.32</u>	2.82	4.44	<u>3.57</u>	4.96	<u>3.72</u>

6 CONCLUSION

We presented CrossEdit, an omni-modal editing model that edits real movie scenes from free-form instructions zero-shot, enabled by two complementary techniques: cross-modal transfer of instruction

following from synthetic audio editing data, and AV masked reconstruction as a self-supervised editing task. We also introduced CrossEditBench and the AV-FES metric for evaluating instruction-based AV movie scene editing. CrossEdit outperforms cascaded, sampling-based, agentic, and SOTA end-to-end models on these tasks while maintaining or improving performance on lip-synced speech, image, and video editing.

AI USE STATEMENT

In this work, we used generative AI tools to generate synthetic datasets, refine hypotheses, design or provide feedback on research methodology or experiments, implement methods, support qualitative and thematic data analysis, and to interpret results.

We have not used generative AI tools to help develop theoretical models or conceptual frameworks, formulate mathematical claims, provide critical ingredients for proving mathematical claims, assist in the writing of proofs, assist with translation, or clean and reformat dataset.

Additionally, we used generative AI tools to create or modify scientific figures or images, create or edit software code, create artifacts, draft parts of a research paper, search for information, edit a research paper to improve readability, and identify relevant literature.

We have reviewed all AI-assisted work. LLM-generated code, figures, and writing were verified and tested (if applicable) for correctness. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This research can help reduce the barrier to entry for creative uses of multimedia, as professional video and audio editing software can be complex to use. It can also reduce the cost of media production, as editing can be used in place of reshoots. On the other hand, such technologies can affect the labor market or even be used for malicious causes. For example, the lip-synced speech editing capabilities of CrossEdit can often result in outputs that are indistinguishable from real videos, which can then be used for fraud, impersonation, or libel. We condemn any use of our research for these causes. Due to the risks of this technology, we do not plan on open-sourcing any code or pre-trained checkpoints.

While most of our evaluations are conducted on data that reflects real-world use cases of multi-media editing models (interviews, movie clips, etc.), it does not consider all possible scenarios. For example, we did not test our models on noisy home-made videos with reverberant audio, as those are difficult to obtain annotations for and score. We also only trained and tested the model on short video clips up to 10 seconds long. While it could theoretically work since it was trained on audio up to 30 seconds long, long-form AV editing remains an open challenging task.

REPRODUCIBILITY STATEMENT

Due to the risks of this technology for making realistic deepfakes and the proprietary nature of the code and training data, the model checkpoints will not be publicly released in their current form. To support reproduction, we describe every component needed to re-implement and evaluate CrossEdit:

- **Model:** the formulation, flow-matching objective, and architecture (Section 3.1, Appendix A), and classifier-free guidance and sampling hyperparameters (Appendix B.3).
- **Training data:** the lip-synced speech editing mining pipeline and its yield (Section 3.2), audio editing data synthesis (Section 3.3), and AV masked reconstruction (Section 3.4, Appendix B.1).
- **Training:** pre-training tasks, SFT data mixture, optimizer, learning rate, and compute (Table 1, Appendix B.2, Table 7).
- **Evaluation:** CrossEditBench construction (Section 4.1, Appendix C.1); the AV-FES definition, including edge cases (Section 4.2, Appendix C.3); judge models, rubric, and agreement across judges (Figure 8, Appendix C.4); the human evaluation protocol (Appendix C.2); and the lip-synced speech editing and audio editing test sets and metrics (Section 5.1, Appendices C.6 and D).

We will release the annotated CrossEditBench evaluation set, annotator instructions, judge prompts, and model outputs on the demo page.

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A ARCHITECTURE

CrossEdit consists of frozen modality-specific tokenizers and a latent diffusion backbone (Rombach et al., 2022) (Figure 2). Each modality is tokenized by a separate pre-trained VAE (Kingma & Welling, 2014):

- **Audio:** a DAC-VAE (Polyak et al., 2024; Kumar et al., 2023; Seetharaman & Kumar, 2026) that encodes 48kHz stereo audio into 256-dimensional continuous latents at 40Hz, pre-trained on over 100K hours of speech, sound effects, and music.
- **Video:** a 3D VAE similar to that of Wan et al. (2025). Although it supports variable resolutions and frame rates, we normalize all videos to 540p at 24 FPS, yielding 48-channel latents of 60×34 at 6 FPS.
- **Images:** treated as single-frame video, yielding one 48-channel 60×34 latent frame for a 540p input.

Following Qwen-Image (Wu et al., 2025a), text, image, and video inputs are additionally encoded into semantic embeddings by a pre-trained Qwen3-VL (Bai et al., 2025b). The backbone is therefore conditioned on clean Qwen3-VL embeddings and clean VAE latents of the input, and denoises the VAE latents of the output. It is a DiT (Peebles & Xie, 2023) with MoE layers (Shazeer et al., 2017), and we use Complete-muE (Peng et al., 2026) to transfer hyperparameters from small-scale experiments.

B TRAINING DETAILS

B.1 AUDIO-VISUAL MASKED RECONSTRUCTION

We frame AV masked reconstruction as a self-supervised AV2AV editing task (Figure 4). Each AV sample is assigned one of three masking schemes at random:

- **Temporal:** a random span of video frames is masked, and with 50% probability the corresponding audio is zeroed out.
- **Spatial:** we sample $s \sim \mathcal{U}\{0, S\}$ and apply s masks of 200×200 px to all video frames. We found $S=12$ to work well for our 540p videos.
- **Spatio-temporal:** both spatial and temporal masking are applied.

All three schemes use the same instruction (“inpaint this video”). The model is conditioned on the masked clip as its input Y and must generate the clean, unmasked clip as its output $\hat{Y}$.

Table 7: **SFT data mixture by total tokens and samples per batch.** Video-to-video and lip-synced speech editing samples are replaced by AV masked reconstruction (Section 3.4) with 50% probability.

Task	Tokens	Batch Samples
<i>Audio</i>		
Synthesized Audio Editing	0.5B	64
Simulated Audio Editing	2.5B	208
Text-to-Audio	0.3B	48
Speech Editing	1.5B	96
<i>Video</i>		
Text to Video	1.3B	4
Video to Video	1.3B	8
Image+Audio to Video	3.2B	10
<i>Audio-Visual</i>		
Text to AV	1.3B	4
Lip-Synced Speech Editing	2.0B	8
<i>Image</i>		
Image to Image	0.2B	12

B.2 PRE-TRAINING AND SUPERVISED FINE-TUNING

Because editing data is scarce, we start from an internal DiT checkpoint pre-trained on large-scale generation and editing data, which we call **CrossEdit-Base**. Its pre-training tasks are:

- **Text-to-X:** text-to-image, text-to-video, and text-to-audio (text-to-speech and text-to-sound effects).
- **Conditional generation:** image-to-video, image-to-AV (talking head generation and animation), audio-to-video, audio-to-AV, and image+audio-to-video (lip-synced talking head generation).
- **Editing:** image-to-image, video-to-video, audio-to-audio (voice cloning only), and AV-to-AV (lip-synced speech editing with the mined pairs of Section 3.2).

We then add the synthetic audio editing (Section 3.3) and AV masked reconstruction (Appendix B.1) data and fine-tune for 10K steps to obtain **CrossEdit**. SFT uses a constant learning rate of $1.2\text{e-}4$, with AdamW (Loshchilov & Hutter, 2019) optimizer states initialized from the pre-trained checkpoint, and takes around 2.5 days on 64 H100 GPUs or 5 days on 64 A100 GPUs. The data mixture is shown in Table 7.

B.3 CLASSIFIER-FREE GUIDANCE AND INFERENCE

During training, we independently replace the input Y and the instruction X with a null token $\emptyset$ at random (Section 3.1). This enables classifier-free guidance (CFG) (Ho & Salimans, 2021) with separate guidance scales for fidelity to the input (λ_Y) and adherence to the instruction (λ_X). The guided velocity is

$$\begin{aligned} \tilde{v}_\theta(\hat{Y}_t, t, Y, X) = & v_\theta(\hat{Y}_t, t, \emptyset, \emptyset) + \lambda_Y \left(v_\theta(\hat{Y}_t, t, Y, \emptyset) - v_\theta(\hat{Y}_t, t, \emptyset, \emptyset) \right) \\ & + \lambda_X \left(v_\theta(\hat{Y}_t, t, Y, X) - v_\theta(\hat{Y}_t, t, Y, \emptyset) \right), \end{aligned} \quad (5)$$

which replaces v_θ when integrating from $t = 1$ to $t = 0$. We use 50 sampling steps with $\lambda_Y = \lambda_X = 5.0$ for video and $\lambda_Y = \lambda_X = 3.0$ for audio unless stated otherwise.

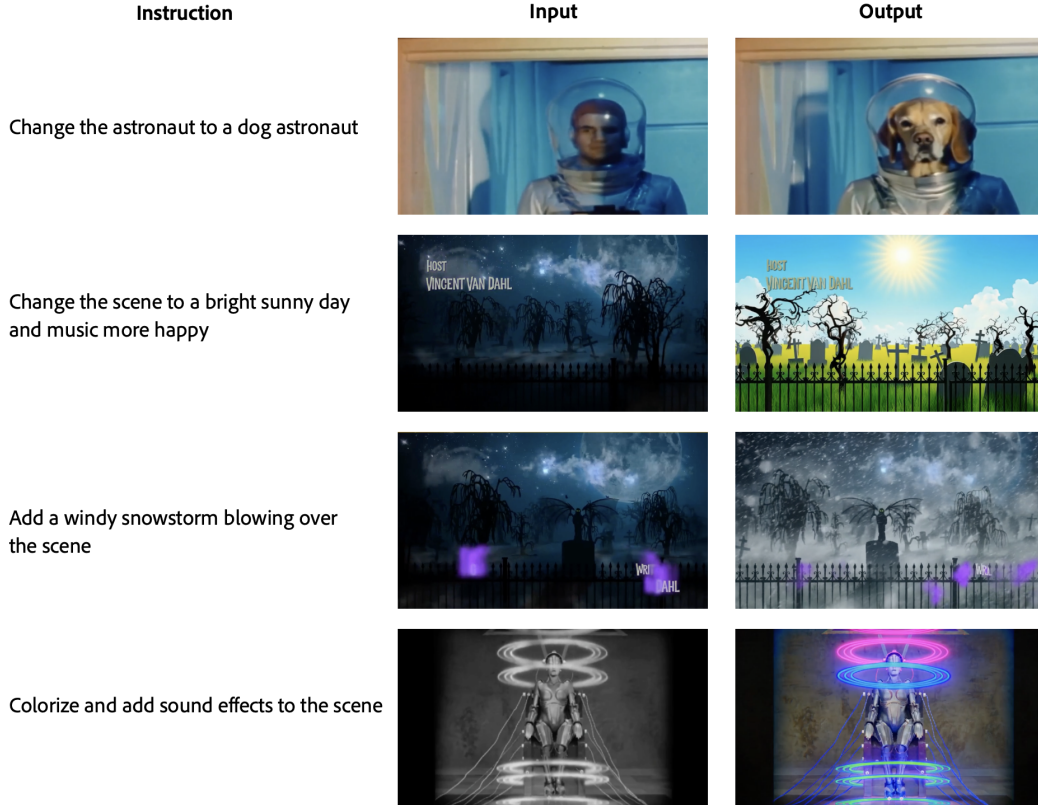


Figure 6: Example AV editing instructions, inputs, and outputs for CrossEditBench/CrossEdit.

C EVALUATION DETAILS

C.1 CROSSEDITBENCH

CrossEditBench evaluates editing of videos with audio on realistic use cases. We annotate 100 clips from OSSSL (Kim et al., 2025), an open-source dataset of public-domain movies originally built for music processing. Each annotation contains the edit instruction (“add a cow and remove the music”), what should change in the audio ([added: cow mooing, removed: music]), and what should change in the video ([added: cow]). We do not use the per-modality annotations in this work but release them to support fine-grained evaluation. As described in Section 4.1, instructions vary in which modality they target and whether they require AV synchrony: an edit may affect both modalities (adding a barking dog), only one modality but in sync with the other (adding a person walking in time with audible footsteps), or only one modality without synchrony (removing the music). Benchmark examples are shown in Figure 1, with additional inputs and outputs in Figures 6 and 7.

C.2 SUBJECTIVE EVALUATION

For AV scene editing, we recruited 100 crowdsourced raters, restricted to English speakers with at least 100 completed tasks and a task approval rate of at least 98%, and required them to wear headphones. Each rater rated 4 clips per system on a 1–5 scale (higher is better) along four axes: visual instruction following, visual consistency, audio instruction following, and audio consistency (instructions in Figure 8). The unedited input is included as a control system to validate rater attention: we exclude the 14 raters who gave it low consistency scores, leaving 86 raters and 344 ratings per system. Full results with 95% confidence intervals, including the + Audio Editing ablation for both tasks, are in Table 3 (AV scene editing) and Table 8 (lip-synced speech editing).

Instruction	Input	Output
Change the cow to a lion		
Ducks quack as they swim across the pond		
Add color and remove the man's hat		
A man steps through the door		

Figure 7: Example AV editing instructions, inputs, and outputs for CrossEditBench/CrossEdit.

Video: You will watch a 2 to 8 second video clip with accompanying audio.

Text Instruction: A text instruction will appear, specifying what changes should be done to the video (e.g., "remove the dog")

Output Video: You will watch another video clip where the algorithm has attempted to change the video and its audio by following the text prompt.

Your task is to evaluate the algorithm's performance along four axes:

Visual Instruction Following: Is the output video correctly changed visually following the text instruction? Do not consider the audio when answering this question.

5 = The video is changed exactly as specified by the text instruction.

1 = The output video does not follow the text instruction at all.

2-4 = The video is changed somewhat according to the text instruction.

Please use the text instruction to guide your judgment.

Visual Consistency: Aside from the specified changes, is the output video visually consistent with the original video? Do not consider the audio when answering this question.

5 = Outside of the specified changes, the output video sounds looks exactly like the original video.

1 = The output video does not resemble the original video at all.

2-4 = The output video resembles the original video, but there are some differences.

Please only consider the consistency with the original video and do not consider the above **Visual Instruction Following** criteria when rating for Consistency.

Audio Instruction Following: Is the output audio correctly changed following the text instruction? Do not consider the video when answering this question.

5 = The audio is changed exactly as specified by the text instruction.

1 = The output audio does not follow the text instruction at all.

2-4 = The audio is changed somewhat according to the text instruction.

Please use the text instruction to guide your judgment.

Audio Consistency: Aside from the specified changes, is the output audio consistent with the original audio? Do not consider the video when answering this question.

5 = Outside of the specified changes, the output audio sounds exactly like the original audio.

1 = The output audio does not resemble the original audio at all.

2-4 = The output audio resembles the original audio, but there are some differences.

Please only consider the consistency with the original audio and do not consider the above **Audio Instruction Following** criteria when rating for Consistency.

Figure 8: Instructions given to human annotators and the LLM Judges

The study was a minimal-risk human evaluation task in which participants viewed videos and provided quality ratings. No sensitive personal information was collected, and participants were informed about the nature of the task and the use of their responses for research purposes. Participation was voluntary, and participants were compensated at approximately \$15 USD per hour. Under our institution's policies, this type of study does not require IRB review or approval.

Table 8: **Full subjective results on IMDB lip-synced speech editing** (1–5, mean $\pm$ 95% confidence interval).

	Audio		Video	
	Ins.	Cons.	Ins.	Cons.
Input	3.29 $\pm$ 0.25	4.39 $\pm$ 0.18	3.29 $\pm$ 0.25	4.50 $\pm$ 0.16
F5-TTS+HuMo	3.30 $\pm$ 0.23	2.97 $\pm$ 0.22	2.60 $\pm$ 0.22	2.10 $\pm$ 0.18
F5-TTS+StableAvatar	3.21 $\pm$ 0.23	2.88 $\pm$ 0.22	2.17 $\pm$ 0.22	1.78 $\pm$ 0.20
CrossEdit-Base	3.98 $\pm$ 0.19	3.81 $\pm$ 0.19	4.05 $\pm$ 0.18	4.22 $\pm$ 0.17
+ Audio Editing	3.86 $\pm$ 0.20	3.82 $\pm$ 0.20	3.99 $\pm$ 0.19	4.35 $\pm$ 0.16
+ AV Masked Reconstruction (CrossEdit)	3.96 $\pm$ 0.20	3.82 $\pm$ 0.20	4.02 $\pm$ 0.19	4.42 $\pm$ 0.15

C.3 AV-FES DETAILS

Motivation. Instruction following and consistency cannot individually measure editing performance well. A system that returns its input untouched scores perfectly on consistency, and a system that regenerates the whole clip can score well on instruction following while discarding everything the instruction did not mention. Neither axis alone answers the question we care about: *did the system make the requested change and nothing else?* Even when both scores are available, it is difficult to weigh increases in one score against drops in the other. AV-FES (Section 4.2) answers it with a single score.

Zero headroom. In the rare case that $\text{Ins}_m(Y) = 5$, the judge deems the instruction already satisfied for modality m , and the gain in Equation 2 is undefined. The correct action is then to leave the modality alone, so we set $g_m(\hat{Y}) = \mathbf{1}[\text{Cons}_m(\hat{Y}) \geq 4]$.

Why the harmonic mean. As in the F_1 score, g_m plays the role of recall (how much of the requested change was made) and c_m that of precision (how much of the output is free of unrequested change). The harmonic mean in Equation 3 is high only when *both* are high: an untouched copy scores 0 because $g_m = 0$, and a full re-synthesis that happens to satisfy the instruction is pulled down by its low c_m . When both terms vanish, we set $\text{FES}_m(\hat{Y}) = 0$.

Range and ceiling. AV-FES is bounded in $[0, 1]$, is zero for the unedited input by construction, and uses the same judge ratings as the Ins. and Cons. subscores in Table 2. In practice, the Gemini judge awards at most 4 on consistency even for byte-identical clips (cf. the input’s consistency scores in Table 2), so c_m saturates near 0.75 and the empirical ceiling of AV-FES for a flawless editor is about 0.86 rather than 1. This compresses the scale uniformly and does not affect rankings.

Single-modality instructions. When an instruction concerns only one modality, the correct behavior in the other modality is to leave it unchanged. AV-FES therefore rewards edits where they were requested and, through c_m , penalizes collateral changes where they were not.

C.4 ROBUSTNESS TO THE CHOICE OF JUDGE

To test whether our conclusions depend on the LLM judge, we re-score CrossEditBench outputs with Qwen3-Omni (Xu et al., 2025) using the same rubric and protocol as the Gemini 3.1 Pro results in Table 2 (100 clips per system). Results for the systems in this paper are in Table 11. To measure agreement between the judges, we compute system-level correlations over a larger pool of 11 systems, including development checkpoints not reported elsewhere in the paper and the unedited input (Table 12). Because the input scores 0 under both judges by construction, we also report correlations without it (10 systems). The judges agree strongly on system rankings in both cases (AV-FES Pearson 0.87, Spearman 0.80 without the input), and both rank CrossEdit first and CrossEdit-Base second among the reported systems. Agreement is highest for video and lower for audio. Qwen3-Omni is considerably more lenient, however: its scores span a range roughly two to three times wider than Gemini’s, and it awards full consistency to the unedited input. AV-FES values should therefore be compared only within a single judge, not across judges.

We also compare each judge against the human ratings of Table 3, which cover a subset of systems: 4 shared with Gemini (the two LucyEdit cascades, CrossEdit-Base, and CrossEdit) and 5 shared with Qwen3-Omni (additionally the + Audio Editing ablation). Both judges rank systems much as humans do (Table 13), with the largest disagreement on Qwen3-Omni’s visual consistency, where it rates CrossEdit above the cascades while humans do not. With so few systems, these correlations are coarse, and a single system can change them substantially.

C.5 CONFIDENCE INTERVALS FOR LLM-JUDGE SCORES

Tables 9 and 10 report 95% confidence intervals over the 100 CrossEditBench clips for both judges. CrossEdit’s AV-FES exceeds that of every baseline under both judges; its margins over the cascades, AvED, AVI-EDIT, and InstructAV2AV are well outside the intervals, while its margin over MiniMax H3 is smaller but consistent across judges.

Table 9: **CrossEditBench results with Gemini 3.1 Pro as the judge, with 95% confidence intervals** (mean $\pm$ half-width over the 100 clips). Means match Table 2.

Input	Judge axes (1–5)				Faithful Edit Score		
	Vis. Ins.	Vis. Cons.	Aud. Ins.	Aud. Cons.	V-FES $\uparrow$	A-FES $\uparrow$	AV-FES $\uparrow$
Input	2.81 $\pm$ 0.19	3.97 $\pm$ 0.03	2.19 $\pm$ 0.10	3.95 $\pm$ 0.05	–	–	–
LucyEdit + HY-XL	2.88 $\pm$ 0.15	3.76 $\pm$ 0.16	2.79 $\pm$ 0.18	2.27 $\pm$ 0.07	0.12 $\pm$ 0.02	0.19 $\pm$ 0.04	0.16 $\pm$ 0.03
LucyEdit + HY-XXL	2.88 $\pm$ 0.16	3.76 $\pm$ 0.16	2.89 $\pm$ 0.17	2.15 $\pm$ 0.10	0.12 $\pm$ 0.04	0.20 $\pm$ 0.05	0.16 $\pm$ 0.03
VACE + Coherent	3.16 $\pm$ 0.17	3.44 $\pm$ 0.16	3.12 $\pm$ 0.17	2.49 $\pm$ 0.14	0.12 $\pm$ 0.04	0.26 $\pm$ 0.04	0.19 $\pm$ 0.03
AvED	2.81 $\pm$ 0.16	3.20 $\pm$ 0.13	2.49 $\pm$ 0.17	3.93 $\pm$ 0.05	0.09 $\pm$ 0.04	0.14 $\pm$ 0.05	0.12 $\pm$ 0.03
AVI-EDIT	2.42 $\pm$ 0.12	2.35 $\pm$ 0.12	2.93 $\pm$ 0.18	3.65 $\pm$ 0.14	0.08 $\pm$ 0.03	0.25 $\pm$ 0.06	0.17 $\pm$ 0.03
InstructAV2AV	2.85 $\pm$ 0.17	3.03 $\pm$ 0.17	2.85 $\pm$ 0.18	3.20 $\pm$ 0.17	0.15 $\pm$ 0.05	0.24 $\pm$ 0.05	0.20 $\pm$ 0.04
MiniMax H3	3.14 $\pm$ 0.16	3.38 $\pm$ 0.15	2.77 $\pm$ 0.18	3.70 $\pm$ 0.11	0.24 $\pm$ 0.06	0.22 $\pm$ 0.06	0.23 $\pm$ 0.04
CrossEdit-Base	2.88 $\pm$ 0.14	2.00 $\pm$ 0.12	3.15 $\pm$ 0.18	3.52 $\pm$ 0.18	0.20 $\pm$ 0.04	0.31 $\pm$ 0.05	0.26 $\pm$ 0.04
CrossEdit	3.59 $\pm$ 0.13	2.81 $\pm$ 0.16	3.70 $\pm$ 0.18	3.84 $\pm$ 0.17	0.26 $\pm$ 0.06	0.33 $\pm$ 0.06	0.30 $\pm$ 0.04

Table 10: **CrossEditBench results with Qwen3-Omni as the judge, with 95% confidence intervals** (mean $\pm$ half-width over the 100 clips). Means match Table 11.

Input	Judge axes (1–5)				Faithful Edit Score		
	Vis. Ins.	Vis. Cons.	Aud. Ins.	Aud. Cons.	V-FES $\uparrow$	A-FES $\uparrow$	AV-FES $\uparrow$
Input	2.69 $\pm$ 0.39	5.00 $\pm$ 0.00	3.32 $\pm$ 0.40	4.40 $\pm$ 0.28	–	–	–
LucyEdit + HY-XL	2.68 $\pm$ 0.38	4.10 $\pm$ 0.29	3.44 $\pm$ 0.38	2.47 $\pm$ 0.36	0.16 $\pm$ 0.09	0.19 $\pm$ 0.12	0.17 $\pm$ 0.07
LucyEdit + HY-XXL	3.88 $\pm$ 0.35	4.18 $\pm$ 0.25	3.84 $\pm$ 0.34	2.55 $\pm$ 0.38	0.50 $\pm$ 0.11	0.28 $\pm$ 0.13	0.41 $\pm$ 0.09
VACE + Coherent	3.85 $\pm$ 0.34	4.20 $\pm$ 0.24	3.73 $\pm$ 0.38	3.03 $\pm$ 0.38	0.33 $\pm$ 0.10	0.15 $\pm$ 0.10	0.25 $\pm$ 0.07
AvED	3.70 $\pm$ 0.37	4.48 $\pm$ 0.18	3.68 $\pm$ 0.36	4.23 $\pm$ 0.30	0.46 $\pm$ 0.12	0.26 $\pm$ 0.13	0.37 $\pm$ 0.09
AVI-EDIT	3.19 $\pm$ 0.39	2.59 $\pm$ 0.33	4.32 $\pm$ 0.30	3.02 $\pm$ 0.39	0.39 $\pm$ 0.10	0.33 $\pm$ 0.13	0.36 $\pm$ 0.08
InstructAV2AV	3.77 $\pm$ 0.36	4.25 $\pm$ 0.21	4.44 $\pm$ 0.28	3.79 $\pm$ 0.35	0.54 $\pm$ 0.12	0.58 $\pm$ 0.15	0.55 $\pm$ 0.09
MiniMax H3	4.19 $\pm$ 0.30	4.68 $\pm$ 0.14	4.13 $\pm$ 0.32	4.13 $\pm$ 0.33	0.69 $\pm$ 0.11	0.55 $\pm$ 0.14	0.63 $\pm$ 0.09
CrossEdit-Base	4.16 $\pm$ 0.32	3.10 $\pm$ 0.29	4.60 $\pm$ 0.24	3.03 $\pm$ 0.36	0.55 $\pm$ 0.09	0.41 $\pm$ 0.13	0.49 $\pm$ 0.08
CrossEdit	4.61 $\pm$ 0.23	4.44 $\pm$ 0.20	4.72 $\pm$ 0.20	4.36 $\pm$ 0.27	0.80 $\pm$ 0.08	0.74 $\pm$ 0.12	0.77 $\pm$ 0.07

C.6 LIP-SYNCD SPEECH EDITING

Lip-synched speech editing is the task of generating new speech and matching lip movements from an instruction and a reference AV clip; it is the audio-visual counterpart of voice cloning (Figure 2). Our IMDB test set consists of 600 two-second clips of different celebrities talking. The videos are shot in varied environments and often include challenging conditions such as multiple visible people (with only one speaker), moving arms and hands, and camera panning. Example inputs and outputs are on our demo page.

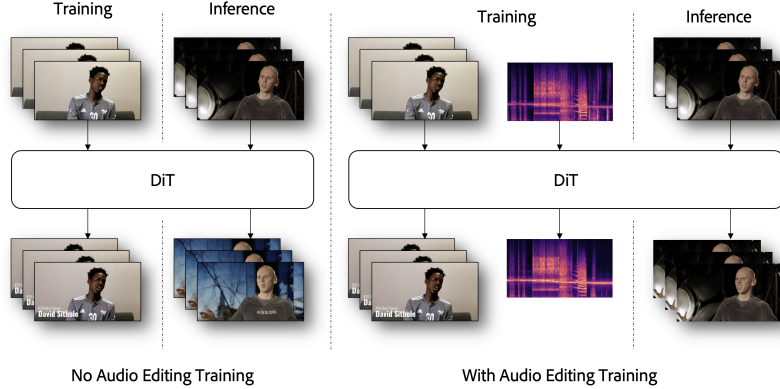


Figure 9: **Synthetic audio editing data reinforces visual consistency.** Mined AV pairs are not perfectly consistent (here, the target has a text overlay and different lighting), causing the model to hallucinate unwanted changes (left). Our method corrects this (right).

Table 11: **CrossEditBench results with Qwen3-Omni as the judge.** Same rubric and protocol as Table 2. AV-FES|edit averages FES_m only over clips whose instruction targets modality *m*. No-gain: number of clips (out of 100) with zero instruction gain.

	Judge axes (1–5)				V-FES ↑	A-FES ↑	AV-FES ↑	AV-FES edit ↑	No-gain ↓
	Vis. Ins.	Vis. Cons.	Aud. Ins.	Aud. Cons.					
Input	2.69	5.00	3.32	4.40	–	–	–	–	–
LucyEdit + HY-XL	2.68	4.10	3.44	2.47	0.16	0.19	0.17	0.17	75
LucyEdit + HY-XXL	3.88	4.18	3.84	2.55	0.50	0.28	0.41	0.38	55
VACE + Coherent	3.85	4.20	3.73	3.03	0.33	0.15	0.25	0.23	60
AvED	3.70	4.48	3.68	4.23	0.46	0.26	0.37	0.35	62
AVI-EDIT	3.19	2.59	4.32	3.02	0.39	0.33	0.36	0.40	53
MiniMax-H3	4.19	4.68	4.13	4.13	0.69	0.55	0.63	0.63	48
InstructAV2AV	3.77	4.25	4.44	3.79	0.54	0.58	0.55	0.56	50
CrossEdit-Base	4.16	3.10	4.60	3.03	0.55	0.41	0.49	0.50	38
CrossEdit	4.61	4.44	4.72	4.36	0.80	0.74	0.77	0.75	34

Table 12: **System-level agreement between Gemini 3.1 Pro and Qwen3-Omni judges.** Computed over 13 systems, including development checkpoints not reported in the main paper and the unedited input, which scores 0 under both judges by construction. The last column excludes the input (12 systems). AV-FES|edit averages FES_m only over clips whose instruction targets modality *m*. Mean |diff| is in score units.

Score	All systems (<i>n</i> =13)					w/o input (<i>n</i> =12)
	Pearson	Spearman	Mean diff	Gemini range	Qwen range	Pearson / Spearman
V-FES	0.96	0.95	0.356	0.00–0.29	0.00–0.80	0.95 / 0.93
A-FES	0.74	0.70	0.185	0.00–0.38	0.00–0.74	0.60 / 0.62
AV-FES	0.90	0.90	0.270	0.00–0.34	0.00–0.77	0.85 / 0.87
AV-FES edit	0.89	0.87	0.215	0.00–0.45	0.00–0.75	0.83 / 0.83

D AUDIO EDITING RESULTS

Setup: We evaluate on three test sets:

- **StoryGenEval** (Chen et al., 2026): complex synthetic scenes generated by a tool-calling LLM agent (similar to WavJourney (Liu et al., 2025)) that uses generative audio models as tools. Instructions are free-form and open-vocabulary.
- **AudioCapsEdit** (Chen et al., 2026): real-world scenes from AudioCaps (Kim et al., 2019) with synthesized edit instructions, testing whether models trained on synthetic data generalize to real recordings. There is no ground-truth output.
- **Movie Separation:** an internal set of professionally mixed movie trailers for instruction-based source separation. The model must extract or remove a list of categories (ambience, music, dialogue, and sound effects). There is no ground-truth output.

Our primary metrics are Δ FLAM (consistency with the input) and editFLAM (instruction following), which Chen et al. (2026) show correlate better with human judgment than FAD (Kilgour et al., 2019) and CLAPScore (Wu et al., 2023b). On Movie Separation, we also report SAM-Audio-Judge (Wang et al., 2026), averaged over judging categories. We compare to AudioChat (Chen et al., 2026), a SOTA instruction-based audio editing model, and SAM-Audio Large (Shi et al., 2025), a SOTA instruction-based source separation model. Since SAM-Audio only supports sound removal, we invert extraction instructions before passing them to it (“extract the dialogue” → “remove the music, ambience, and sound effects”). CrossEdit-Base is omitted as it was not trained for complex audio editing.

Results: CrossEdit outperforms prior work on all three test sets (Table 14). The largest gain is on AudioCapsEdit (40.3 vs. 15.5 editFLAM over AudioChat), which we attribute to our large-scale synthetic mixtures. CrossEdit also surpasses SAM-Audio on source separation, suggesting that our synthetic data strategy is more effective than its pseudo-labeling of real mixtures.

Table 13: **System-level agreement between LLM judges and human ratings** (Pearson / Spearman) over the systems shared with Table 3: $n=4$ for Gemini 3.1 Pro and $n=5$ for Qwen3-Omni.

Score	Gemini 3.1 Pro	Qwen3-Omni
Vis. Ins.	0.93 / 1.00	0.82 / 1.00
Vis. Cons.	0.99 / 1.00	0.71 / 0.60
Aud. Ins.	0.71 / 0.50	0.94 / 0.97
Aud. Cons.	1.00 / 0.90	0.93 / 1.00
V-FES	0.92 / 1.00	0.90 / 1.00
A-FES	0.84 / 0.50	0.99 / 1.00
AV-FES	0.92 / 1.00	0.96 / 1.00
AV-FES edit	0.94 / 0.90	0.97 / 1.00

Table 14: **Audio editing results.** StoryGenEval tests complex instruction following, AudioCapsEdit real-world editing, and Movie Separation instruction-based source separation. SAM-J: SAM-Audio-Judge.

	StoryGenEval		AudioCapsEdit editFLAM $\uparrow$	Movie Separation		
	Δ FLAM $\downarrow$	editFLAM $\uparrow$		Δ FLAM $\downarrow$	editFLAM $\uparrow$	SAM-J $\uparrow$
SAM-Audio	-	-	-	18.97	4.12	2.96
AudioChat	11.7	18.6	15.5	-	-	-
CrossEdit	6.64	23.1	40.3	7.08	4.29	3.41